Illustration

CPR Coach

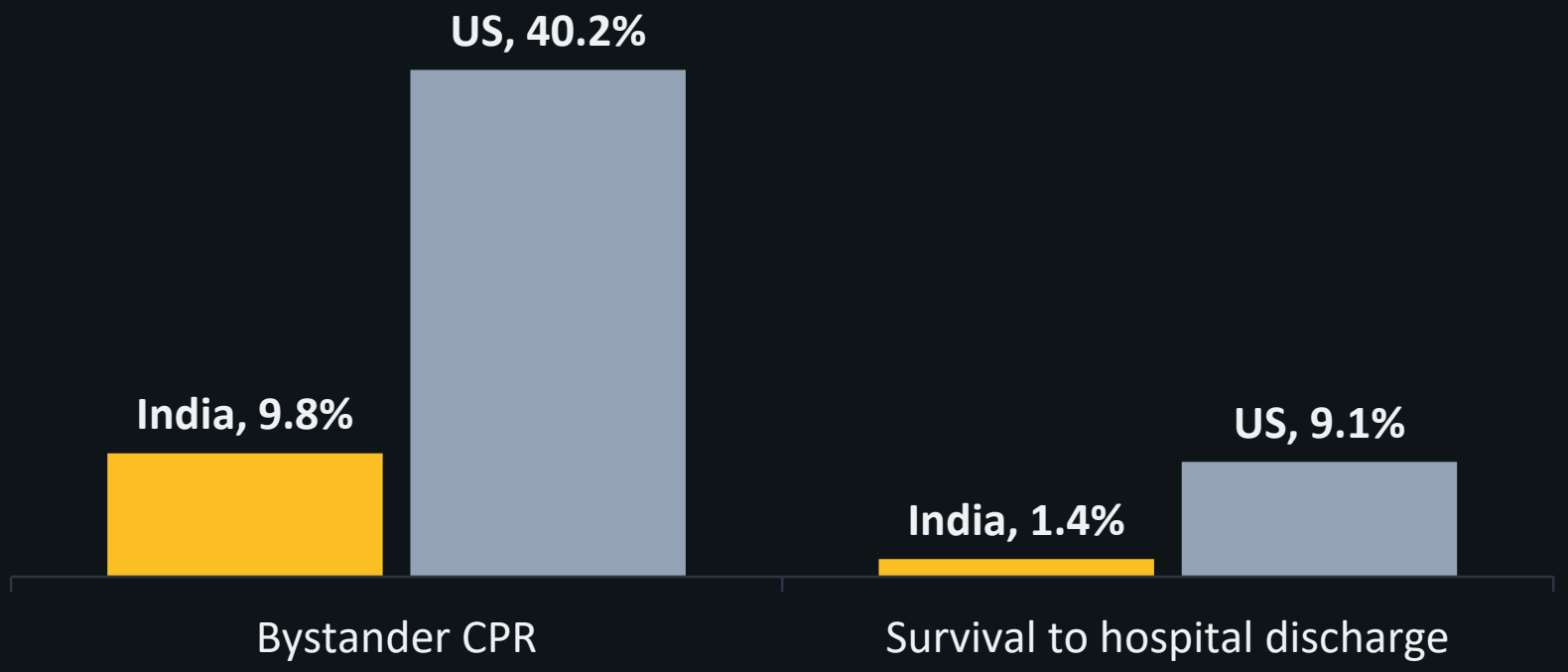
A phone that watches your CPR and tells you what to fix.



Team Zero Flux Vivek · Pratham · Prabal · Moksh

ANVATHON 2026 · OPEN INNOVATION

India's bystander CPR rate is a quarter of the US rate



India
one hospital, 205 patients

US
national registry

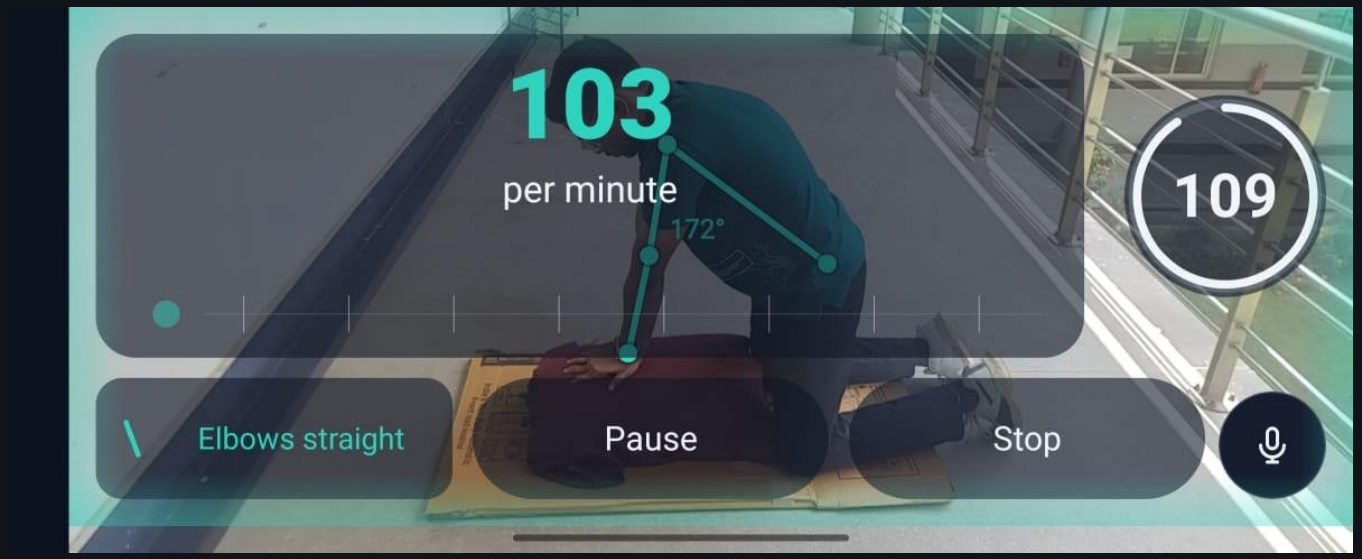
Fewer people nearby start CPR, and fewer patients leave hospital alive.

Karnataka is moving: in July 2025 the Health Minister announced CPR training for school and college students, teachers and gym trainers.

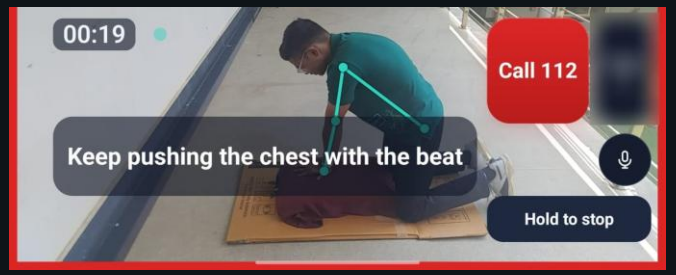
India: Bhat et al., Indian Heart Journal 2021, one tertiary hospital, 205 patients.
Karnataka: NewKerala, 10 July 2025.
The two sources are not measured the same way: one hospital vs a national registry.

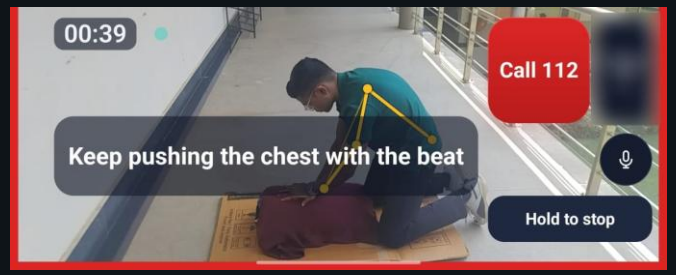
US: American Heart Association, Heart Disease and Stroke Statistics 2023.
Bystander CPR: over 356,000 out-of-hospital arrests. Survival: 2021 US data, EMS-treated adult non-traumatic arrests.

It watches your arms and says what to fix



Practise: 103 pushes a minute, elbows straight





Emergency: joints turn amber when posture needs fixing

Watch the camera finds shoulder, elbow and wrist

Count push rate, elbow angle and pauses

Speak one correction at a time, out loud



All you need: a phone and a home-made manikin.

Measures rate, elbows and shoulders over hands. Can't measure depth, force or hand placement.

Coaches in 10 languages, fully offline.

Illustration

Built in 24 hours: 596 commits

FROM

Our first real run on a phone crashed mid-session, and the recording was lost.

TO

A working prototype: the crash fixed, camera load capped, and heat and battery logged on every run.

399

tests passing, 0 failing

7

real phone recordings must replay byte for byte

Illustration

Live demo

Next: hand-counted accuracy runs, review by CPR trainers, college pilots.

Not a medical device. In an emergency, call 112 (108 for an ambulance) and start CPR.

Code and test records

`github.com/VivekTiwary-25/Zero_flux`

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